Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device).*

The RS485 Digital IO module is sturdy, low power usage and easy to use.

16 Port DI and 16 Port DO Module: -

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The Soft Setting for Slave ID and Baud rate, parity bit, stop bit.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm and 2 Pin 5.08 mm pitch pluggable screw Terminal Block
Dimensions	110 mm L x 110 mm B x 50mm H
Power	Input Power – 15-40V DC or 24 V AC/DC Typical – 12V DC @ 150mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	RoHS, APPROVED CE APPROVED

Digital Output

Channels	16
Output Type	P-Channel MOSFET
Channel Voltage	12 VDC
Channel Current	20A
Channel Power	240 W (max)
Operating Time	2.8ns.
Turn Off delay Time	33.5ns.
Operating Temperature	-55℃~+150℃

DI Inputs –

Channels	16
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic ‘0’ = <1 VDC, Logic ‘1’ = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 Msec

Additional Features -

All inputs and communication port isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
400V isolation.
CRC Error check.
No configuration needed on the IO board

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	DI 1(With & Without Potential)	00001	0X02	RS485	1 Bit Boolean
2	DI 2(With & Without Potential)	00002	0X02	RS485	1 Bit Boolean
3	DI 3(With & Without Potential)	00003	0X02	RS485	1 Bit Boolean
4	DI 4(With & Without Potential)	00004	0X02	RS485	1 Bit Boolean
5	DI 5(With & Without Potential)	00005	0X02	RS485	1 Bit Boolean
6	DI 6(With & Without Potential)	00006	0X02	RS485	1 Bit Boolean
7	DI 7(With & Without Potential)	00007	0X02	RS485	1 Bit Boolean
8	DI 8(With & Without Potential)	00008	0X02	RS485	1 Bit Boolean
9	DI 9(With & Without Potential)	00009	0X02	RS485	1 Bit Boolean
10	DI10(With & Without Potential)	00010	0X02	RS485	1 Bit Boolean
11	DI11(With & Without Potential)	00011	0X02	RS485	1 Bit Boolean
12	DI12(With & Without Potential)	00012	0X02	RS485	1 Bit Boolean
13	DI13(With & Without Potential)	00013	0X02	RS485	1 Bit Boolean
14	DI14(With & Without Potential)	00014	0X02	RS485	1 Bit Boolean
15	DI15(With & Without Potential)	00015	0X02	RS485	1 Bit Boolean
16	DI16(With & Without Potential)	00016	0X02	RS485	1 Bit Boolean
17	DO1	00017	0X05,0X0F	RS485	1 Bit Boolean
18	DO2	00018	0X05,0X0F	RS485	1 Bit Boolean
19	DO3	00019	0X05,0X0F	RS485	1 Bit Boolean
20	DO4	00020	0X05,0X0F	RS485	1 Bit Boolean
21	DO5	00021	0X05,0X0F	RS485	1 Bit Boolean
22	DO6	00022	0X05,0X0F	RS485	1 Bit Boolean
23	DO7	00023	0X05,0X0F	RS485	1 Bit Boolean
24	DO8	00024	0X05,0X0F	RS485	1 Bit Boolean
25	DO9	00025	0X05,0X0F	RS485	1 Bit Boolean
26	DO10	00026	0X05,0X0F	RS485	1 Bit Boolean
27	DO11	00027	0X05,0X0F	RS485	1 Bit Boolean
28	DO12	00028	0X05,0X0F	RS485	1 Bit Boolean

29	**DO13**	00029	0X05,0X0F	RS485	1 Bit Boolean
30	**DO14**	00030	0X05,0X0F	RS485	1 Bit Boolean
31	**DO15**	00031	0X05,0X0F	RS485	1 Bit Boolean
32	**DO16**	00032	0X05,0X0F	RS485	1 Bit Boolean
Soft Setting Configuration in function code 3					
10	**Display Baud Rate (Default:960)**	40011	0X03	RS485	16 Bit Unsigned int
11	**Enter New Baud Rate**	40012	0X03	RS485	16 Bit Unsigned int
12	**Display Slave ID**	40013	0X03	RS485	16 Bit Unsigned int
13	**Enter New Slave ID**	40014	0X03	RS485	16 Bit Unsigned int
14	**Display Parity**	40015	0X03	RS485	16 Bit Unsigned int
15	**Enter New Parity**	40016	0X03	RS485	16 Bit Unsigned int
16	**Display Stop Bit**	40017	0X03	RS485	16 Bit Unsigned int
17	**Enter New Stop Bit.**	40018	0X03	RS485	16 Bit Unsigned int

Configuration Settings -

Communication Speed	9600 - 115200(Soft Setting)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Soft Setting
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31
Function code DI	0x02 Read Discrete Input
DI Register Address	0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15.
Function Code Soft Serial	0x03 Read Holding Registers
Soft Serial Register Address	11,12,13,14,15,16,17,18.

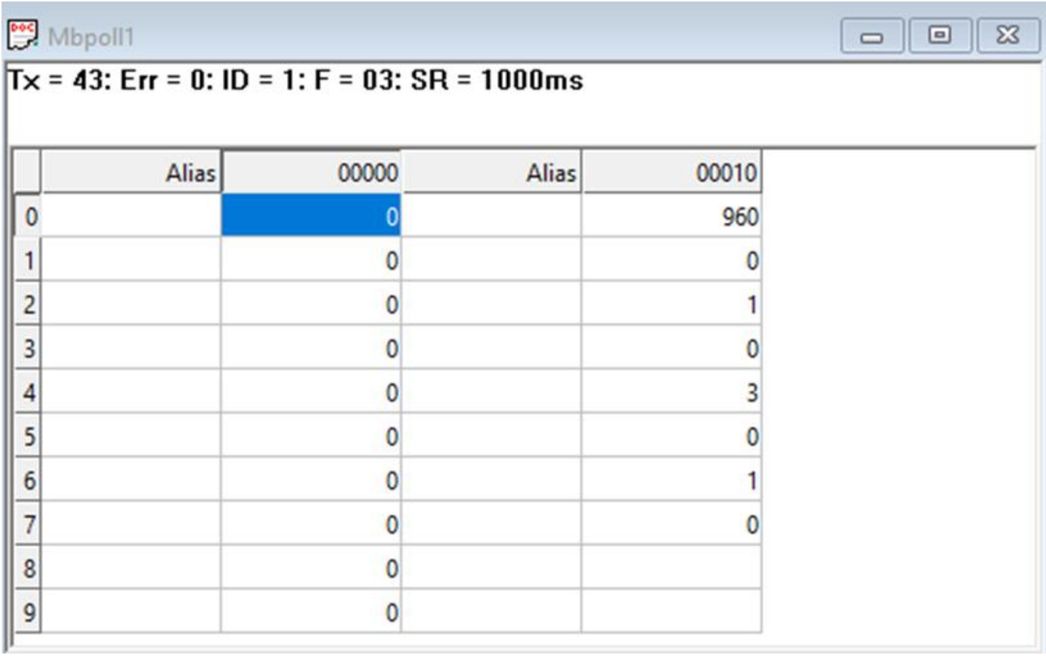
Note: -

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

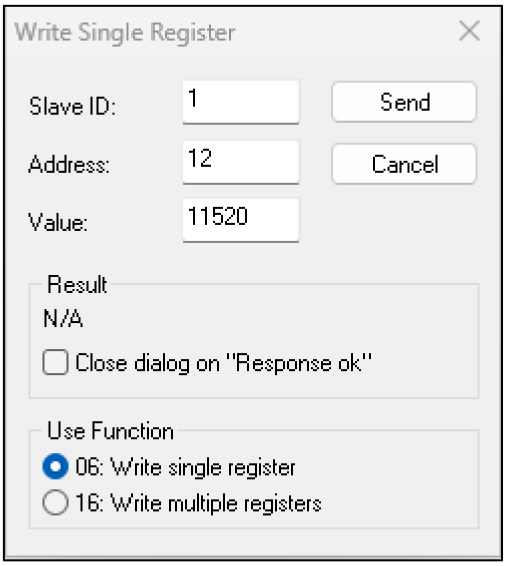
BAUD RATE and SLAVE ID DESCRIPTION:

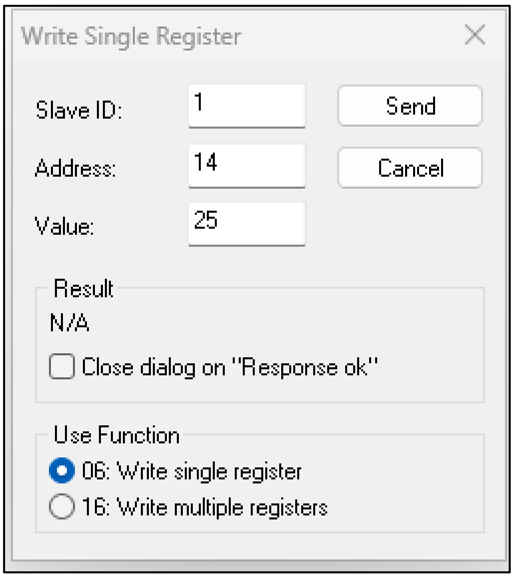


	Alias	00000	Alias	00010
0		0		960
1		0		0
2		0		1
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

- **As you can see in the image above, the resistor 40011 – 960 represents the present bund rate**
- **(9600) of the board and 40013 – 1 represents the present slave id of the board.**

- **To change the baud rate and slave id you have to enter a value on the 40012 resistor for the baud rate and 40014 for the slave id as shown in the image below**





Mbpoll1

Tx = 21: Err = 0: ID = 1: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		960
1		0		11520
2		0		1
3		0		25
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in the image above I set 115200 baud rate and slave ID 25.

Note:

1. **If you want to enter the baud rate 9600/19200/38400/57600/115200etc.
   Enter it like this 960/1920/5760/11520etc.**
2. **After that press the reset button or reboot the system.**
3. **Then you will get the present value of 11520 at 40011 baud rate(115200) and 25 at 40013 slave id.**
4. **You can Change Parity Bit , stop bit Of the device with same way.**

Mbpoll1

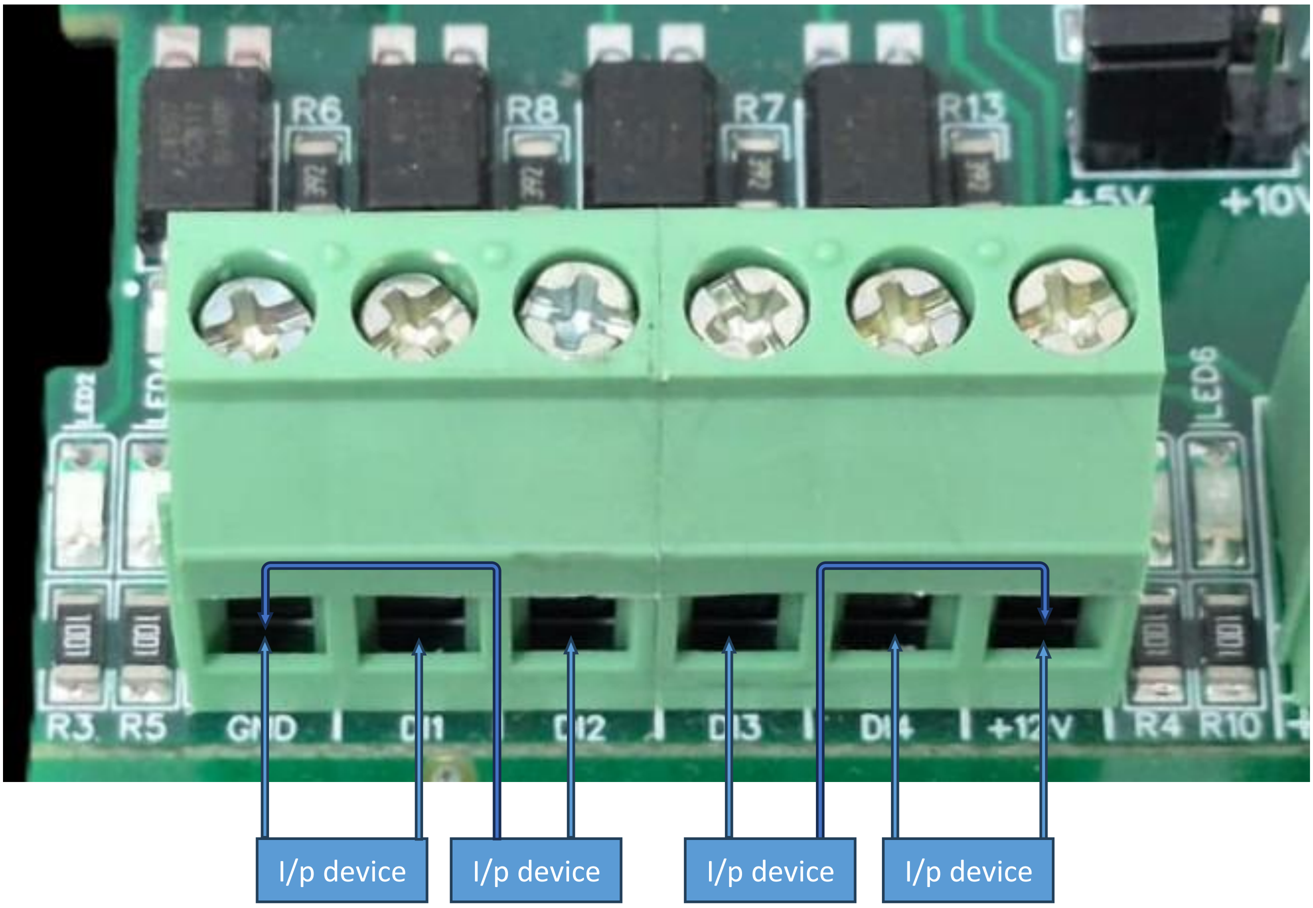
Tx = 12: Err = 5: ID = 25: F = 03: SR = 1000ms

	Alias	00000	Alias	00010
0		0		11520
1		0		0
2		0		25
3		0		0
4		0		3
5		0		0
6		0		1
7		0		0
8		0		
9		0		

As you can see in image, baud rate and slave id are changed.

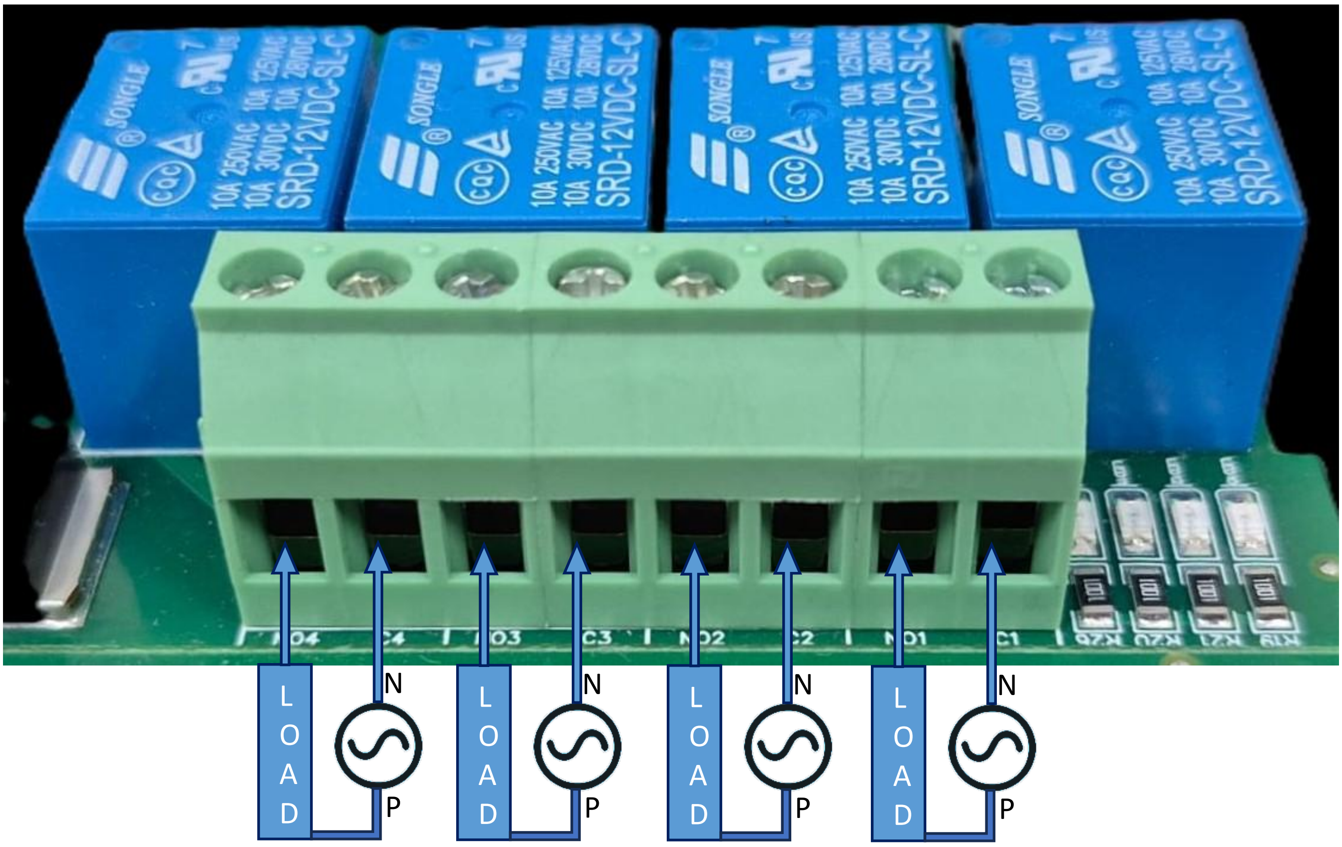
- **To reset Modbus settings to default, remove the reset jumper and press the reset button. The device will enter default mode.**
- **Configuration i.e. Baud rate - 9600, Stop bit - 1, Parity - None, e.g. 12v & DI1+ve terminal.**

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
  - **GND:** Common Ground reference.
  - **DI1 – DI4:** Individual Digital Input channels.
  - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
  - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
  - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

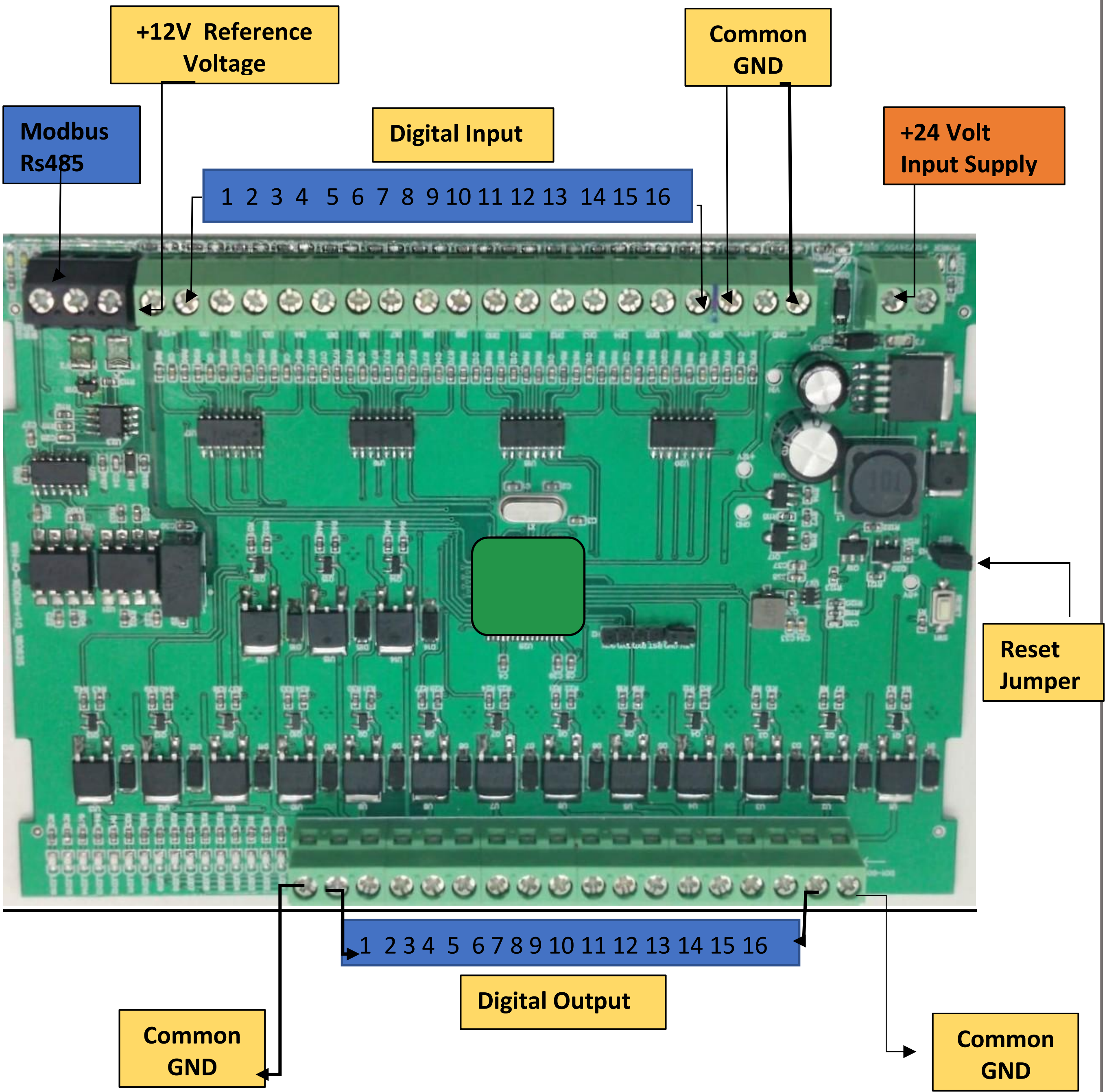
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C is internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO remains open**, and the load stays OFF.
- **Load Wiring Connections:**
  - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
  - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

  <u>**Note:**</u> Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or 10A @ 125VAC) or **10A @ 30VDC** (or 10A @ 28VDC).

WIN – IO –16DDM CE



Contact us: -

Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.
Email – Sales@wittelb.com
Support: support@wittelb.raiseaticket.com / +91 94038 92805